NMC 2010 O-LEVEL PAPER 1 QUESTIONS

SECTION A

Qn.1. What is the value of;

$$\log_5\left(\frac{(125)(625)}{25}\right)?$$

Qn.2. Two non-zero real numbers, a and b satisfy the relationship $ab = a - b$, find the value of

$$\frac{a}{b} + \frac{b}{a} - ab?$$

Qn.3. An entertainment group consists of boys and girls. When 15 girls leave the group, the ratio of boys to girls remaining is $2:1$. Later if 45 boys leave, the ratio of boys to girls changes to $1:5$. Find the number of girls in the group at the beginning?

Qn.4. The total in-store price for an appliance is UGX $200,000$. A television commercial advertises the same product for three easy payments each more than a quarter of the in-store price by UGX $5,000$ and a one time transportation and handling charge of UGX $15,000$. What percentage of the total in-store price is saved by buying the appliance from the television advertiser?

Qn.5. At the end of 1994, Walter was half as old as his grand father. The sum of the years in which they were born is 3838. How old will Walter be at the end of 2010?

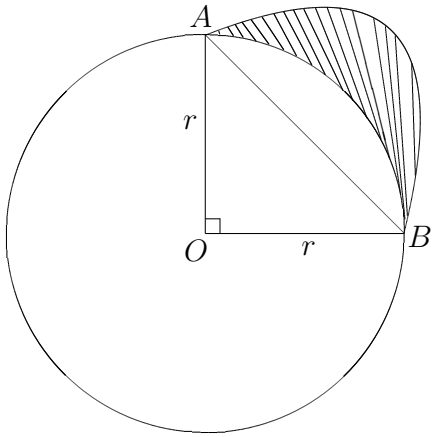
Qn.6. Each day, Jesse ate 20% of the mangoes that were in his fridge at the beginning of that day, at the end of the second day, 32 remained. How many mangoes were in the fridge originally?

Qn.7. Find the sum of all prime numbers between 1 and 100 that are both 1 greater than a multiple of 4 and 1 less than a multiple of 5?

Qn.8. Find the area of the largest triangle that can be inscribed in a semi-circle whose radius is r?

Qn.9. A man buys a house for $20,000,000$ UGX and rents it out. He puts $12\frac{1}{2}\%$ of each months rent aside for repairs and up keep, pays $650,000$ UGX a year in taxes and realizes $5\frac{1}{2}\%$ on his investment. What is the monthly house rent?

Qn.10. The circle below has center O, $\overline{OA}$, and $\overline{OB}$ are radii and $< AOB$ is a right angle. A semi circle is constructed using segment AB as it's diameter as shown above. The shaded portion of the semi circle outside circle O is called a *lune*. What is the ratio of the area of the lune to the area of the triangle?



SECTION B : 10 **marks each**

Qn.11. (a) Okello can do a piece of work in 9 days. Oryem is 50% more efficient than Okello. Find the number of days it takes Oryem to do the same piece of work?

(b) If the width of a rectangle is increased by 10% and the area of the rectangle remains constant. Find the percentage by which the length would have been decreased in order to achieve this?

Qn.12. Two people D and M start at the same time to drive from Kampala to Masaka, 60 miles away. D travels 4 miles an hour slower than M. M reaches Masaka and at once turns back meeting D 12 miles from Masaka. Find the speed of M.

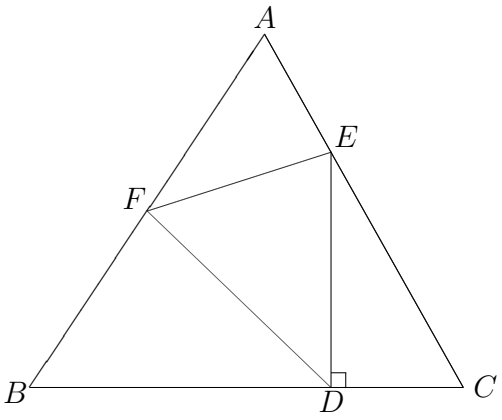
Qn.13. Let $E(n)$ denote the sum of even digits of n for example $E(5681) = 6 + 8 = 14$. Find the value of

$$E(1) + E(2) + E(3) + \ldots + E(100)?$$

Qn.14. The digits $2, 4, 5, 6, 8$ and 9 can be distributed among the lettered squares in the array so that the sum of the entries on each of the rows and columns is the same number K. Identify all the lettered digits and hence find the value of K?

7	a	b	1
c			d
3	e	f	10

Qn.15. Equilateral triangle DEF is inscribed in equilateral triangle ABC as shown below with $\overline{DE} \perp \overline{BC}$. Find the ratio of the area of $\triangle DEF$ to the area of $\triangle ABC$?


A
E
F
B
D
C